

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

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Lab Objectives

- Understand relational database schema design
- Generate SQL queries using AI tools
- Perform CRUD and aggregation operations
- Apply normalization and optimization techniques

Task 1: Student Information System

Schema

```
CREATE TABLE Students(  
    id INT PRIMARY KEY,  
    name VARCHAR(50)  
);
```

```
CREATE TABLE Courses(  
    id INT PRIMARY KEY,  
    course_name VARCHAR(50)  
);
```

```
CREATE TABLE Enrollments(  
    student_id INT,  
    course_id INT  
);
```

Queries

```
INSERT INTO Students VALUES(1, 'Sanju');
```

```
SELECT * FROM Enrollments WHERE student_id = 1;
```

```
SELECT course_id, COUNT(*) FROM Enrollments GROUP BY course_id;
```

Task 2: Hospital Database

```
CREATE TABLE Doctors(id INT PRIMARY KEY, name VARCHAR(50));  
CREATE TABLE Patients(id INT PRIMARY KEY, name VARCHAR(50));  
CREATE TABLE Appointments(id INT, doctor_id INT, patient_id INT);
```

```
SELECT * FROM Appointments WHERE doctor_id = 1;
```

```
SELECT * FROM Patients WHERE id = 1;
```

```
SELECT doctor_id, COUNT(*) FROM Appointments GROUP BY doctor_id;
```

Task 3: Library Database

```
CREATE TABLE Books(id INT, title VARCHAR(50));  
CREATE TABLE Members(id INT, name VARCHAR(50));  
CREATE TABLE Loans(book_id INT, member_id INT);
```

```
SELECT * FROM Loans;
```

```
SELECT * FROM Loans WHERE loan_date > 30;
```

```
SELECT member_id, COUNT(*) FROM Loans GROUP BY member_id;
```

Task 4: E-Commerce Database

```
CREATE TABLE Users(id INT, name VARCHAR(50));
```

```
CREATE TABLE Products(id INT, name VARCHAR(50));
CREATE TABLE Orders(id INT, user_id INT);
CREATE TABLE OrderDetails(order_id INT, product_id INT);

SELECT * FROM Orders WHERE user_id = 1;

SELECT product_id, COUNT(*) FROM OrderDetails GROUP BY product_id;

SELECT SUM(price) FROM Orders;
```

Task 5: University Database

```
CREATE TABLE Students(id INT, name VARCHAR(50));
CREATE TABLE Subjects(id INT, name VARCHAR(50));
CREATE TABLE Results(student_id INT, subject_id INT, marks INT);

INSERT INTO Results VALUES(1, 1, 90);

SELECT subject_id, marks FROM Results WHERE student_id = 1;

SELECT subject_id, AVG(marks) FROM Results GROUP BY subject_id;
```

Conclusion

AI-assisted database design simplifies schema creation and query generation.

Proper normalization and optimized queries ensure efficient data management.